Paper 21.25 - MorphForge Toolkit Supplement

CQE-paper-21.25
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-21.25.md

Paper 21.25 - MorphForge Toolkit Supplement

Purpose

This supplement describes the tools that may be used to reproduce Paper 21. It is not the proof itself. The proof-carrying item is the lossless ribbon codec, the morphonics ledger receipt, and the terminal landing check in Paper 21.

Digital Route

Run the Paper 21 verifier:

`python production/formal-papers/CQE-paper-21/verify_morphforge_ribbon.py`

The expected result is `pass_with_open_obligations`. A valid run writes `morphforge_ribbon_receipt.json` and reports:

- zero chart-codec round-trip mismatches,
- a 1569-state shell-2 ribbon,
- a 1568-step S3 word,
- 494 identity self-loops,
- 1074 non-identity fold steps,
- a morphonics ledger with five passing morphon closure tests,
- three explicit open failure labels,
- a 24-dimensional `Niemeier:E8^3` terminal landing template.

Analog Route

To reproduce the idea by hand, draw a row of adjacent local windows. Mark each window as left, center, and right. For each adjacent pair, record whether the local ordering is unchanged or whether two positions exchange. Unchanged pairs are identity self-loops. Exchanges are folds. The analog record is accepted only when the reader can replay the marks and recover the same ribbon.

The analog route is useful because it exposes the base move: no hidden computer feature is needed to understand the proof. A person can draw the ribbon, mark the folds, and audit whether a later claim is adding information that was not in the initial enumeration.

Hidden-Guess Diagnostic

When training mode is enabled, every non-math validation should hide the diagnostic label until after the agent or reviewer commits to a choice. Paper 21 uses these labels:

- closed ribbon,
- fold event,
- morphon schema closure,
- terminal template placement,
- open bridge,
- invalid promotion.

The answer key is revealed only after the choice. This keeps the diagnostic from becoming a prompt-following exercise and turns each validation into a small ablation.

Boundaries

The toolkit does not prove cross-medium equivalence, material behavior, biological behavior, CAD correctness, a Mandelbrot theorem, or a Leech lattice construction. It supplies the reader and the audit trail those later claims must use.